Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/24/2006
PI Name:	Graham, Michael D. Ph.D.		
Project Title:	Enhancing Performance Of Surface-Based Microfluidic Arrays With Chaotic Flows Of Complex Fluids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	graham@engr.wisc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	
Organization Name:	University of Wisconsin-Madison		
PI Address 1:	Chemical and Biological Engineering, 3010 Engineering Hall		
PI Address 2:	1415 Engineering Dr.		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53706 - 1607	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
Start Date:	07/01/2004	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Abbott, Nicholas (University of Wisconsin - Madison)		
Grant/Contract No.:	NNC05GA38G		
Performance Goal No.:			
Performance Goal Text:			
	Surface-based bioanalytical methods are being investigated for a range of biodetection applications because (1) immobilization of receptors on surfaces permits easy separation of target molecules and non-targeted species, (2) patterning of receptors on surfaces permits highly multiplexed assays to be performed on the same sample, and (3) a variety of surface-based analytical methods exist to detect species bound to surfaces. Many of these applications are central to diagnostic and analytical procedures that may be performed during space missions. The successful development of these methods, however, requires that mass transport of analytes to surfaces be enhanced. Microfluidics is one approach under investigation because it leads to accelerated rates of mass transport and permits assays to be performed with extremely small quantities of reagents. Even when using microfluidic channels, however, it takes a minute or longer for relevant biological species (e.g., proteins or viruses) in aqueous solutions to diffuse the 100 microns that characterize the scale of a typical device. Motivated in part by this observation, this proposal will focus on		

Task Description:	hydrodynamic methods for enhancing rates of transport, and specifically binding to surfaces, in low Reynolds number flows. Issues that will be addressed in the course of the proposed work include (1) shear-induced diffusion in suspensions of particles with fractal structure, which preliminary results show to be significantly enhanced relative to that for suspensions of spheres and (2) predicted synergies between shear-induced diffusion and chaotic mixing, leading to mixing and adsorption enhancement greater than would be obtained by application of each process individually. This will be the first study specifically addressing these issues. Spherical particles as well as robust fractal particles formed by irreversible aggregation of polystyrene beads will be studied in a variety of flow situations. Simple studies of the broadening of an interface between dyed and undyed fluid will be performed. Flow reversal experiments will be used to assess the interaction between shear-induced diffusion and chaotic mixing by determining how these affect the level of irreversibility of the flow. Adsorption experiments will directly determine the extent to which these phenomena enhance transport to surfaces, with specific attention to the degree of synergy between the processes. Particle level simulations will elucidate the dynamics of collisions between fractal particles in shear flow, the fundamental event underlying shear-induced diffusion, and continuum modeling will be used to better understand the interaction between shear-induced diffusion and chaotic flow kinematics. The outcome of the proposed work will be a qualitatively better understanding of two primary phenomena that generate enhanced transport in low Reynolds number flows, and an assessment of their collective effectiveness in improving performance of surface-based microfluidic assays.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No report received this reporting period.
Bibliography Type:	Description: (Last Updated:)